

Notes: He, Qian & Strahan (JF, 2012)
**Are All Ratings Created Equal? The Impact of Issuer Size on the Pricing of
Mortgage-Backed Securities**

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1 Big picture

- **Question.** Are MBS credit ratings equally informative across issuers, or did large issuers receive more inflated ratings from rating agencies during the private-label MBS boom?
- **Core hypothesis.** Under the issuer-pays model, large MBS issuers provide repeated business to rating agencies. This bargaining power may create rating-agency conflicts of interest, especially during boom years when agency revenues are high.
- **Empirical idea.** If investors suspect that large issuers receive more generous ratings, then tranches with the same stated rating should carry higher initial yields when sold by large issuers. If the market fails to fully price this risk, those tranches should perform worse ex post.
- **Setting.** The paper studies privately issued, non-GSE mortgage-backed securities originated from 2000 to 2006 and follows their prices through April 2009.
- **Main comparison.** The key variable is lagged issuer market share. Large issuers are institutions with large shares of the private-label MBS issuance market.
- **Boom-period channel.** The paper interacts issuer market share with a measure of hot MBS market conditions. The interaction is the key test because conflicts of interest should be strongest when issuance volume and rating-agency fee revenue are high.
- **Main yield result.** During boom years, MBS tranches sold by large issuers have higher initial yields than otherwise similar tranches sold by small issuers, even when they have the same rating.
- **AAA result.** The effect appears even among AAA tranches, suggesting that investors treated AAA ratings from large issuers as less credible.
- **Non-AAA result.** Similar effects appear for non-AAA tranches, and ratings-shopping proxies become especially informative in this market.
- **Ex post result.** Securities sold by large issuers during 2004–2006 drop more in price after issuance than securities sold by small issuers.
- **Regulatory-arbitrage result.** MBS sold by banks and thrifts after July 2004 command higher yields, consistent with the idea that capital-regulation incentives affected deal structure and perceived risk.
- **Economic takeaway.** Ratings were not priced as homogeneous signals. Investors demanded higher yields for securities issued by large institutions when conflicts of interest and rating inflation were most plausible.

2 Data and measurement

2.1 Sample construction

- The sample starts with Securities Data Corporation data on privately issued, non-GSE MBS deals.
- The authors manually match tranche-level data from Bloomberg, including ratings, prices, coupons, principal amount, coupon type, weighted average life, and payoff information.
- The issuance window is 2000–2006, and the price-history window extends through April 2009.
- The sample excludes agency MBS because the paper wants to study private-label securities where rating agencies play a central information role.
- The unit of observation in the main regressions is the tranche.

Interpretation. The setting is ideal for testing rating conflicts because private-label MBS deals are complex, repeated, and heavily dependent on ratings.

2.2 Issuer size and market conditions

- **Issuer Share** is the issuer’s lagged annual market share in private-label MBS issuance.
- Figure-based comparisons define **big issuers** as issuers in the top 10% of the issuer market-share distribution in that year.
- Regression specifications use the continuous lagged market share rather than only a big-issuer dummy.
- **HOT** is a year-level measure of MBS market heat: the principal amount issued in a year relative to total issuance across the full 2000–2006 sample.
- Because HOT varies only by year, issuance-year fixed effects absorb its level. The identifying term is $IssuerShare \times HOT$.

Interpretation. The coefficient on $IssuerShare \times HOT$ asks whether large issuers are priced differently specifically when the market is booming and rating-agency conflicts are likely to be more severe.

2.3 Ratings and rating-shopping variables

- Ratings come from Moody’s, S&P, and Fitch.
- AAA tranches are those rated AAA by all agencies that rate the tranche.
- Non-AAA tranches use initial rating category dummies based on a numerical conversion of ratings.
- **One Initial Rating** and **Two Initial Ratings** indicate tranches with only one or two reported ratings rather than three.
- **Rating Disagreement** equals one when multiple reported ratings disagree.
- **Log Number of Tranches** proxies for deal complexity.

Interpretation. Fewer ratings and rating disagreement are observable shadows of potential ratings shopping. The issuer may have solicited preliminary views, purchased the favorable rating, and hidden unfavorable information from investors.

2.4 Deal, tranche, collateral, and issuer controls

- Tranche controls include log principal amount and log weighted average life.
- Deal-structure controls include Fraction AAA for AAA tests and Level of Subordination for non-AAA tests.
- Collateral controls include the fraction of collateral in troubled states and a Herfindahl index of collateral geographic concentration.
- Issuer controls include issuer rating and indicators for bank/thrift issuers.
- A key regulatory variable is $Bank/Thrift \times Post\ July\ 2004$.
- All regressions include coupon-type intercepts and issuance-year fixed effects.
- Standard errors are double clustered by issuer and cohort year.

Interpretation. These controls are designed to separate issuer-size pricing from obvious credit-risk, deal-design, collateral-quality, and issuer-quality differences.

2.5 Dependent variables

The ex ante pricing outcome is

$$\log(YieldSpread_{i,j,t}),$$

where the yield spread is defined as a fixed markup over the reference rate for floating-rate tranches or the initial coupon less a matched Treasury yield for fixed/variable coupon tranches.

The ex post performance outcome is

$$PriceChange_{i,j,t},$$

the percentage price change from the first month after issuance to April 2009, payoff date, or the last available monthly price.

Interpretation. The first outcome asks what investors priced at issuance. The second asks whether large-issuer tranches subsequently performed worse when the market collapsed.

3 Models and empirical specifications

3.1 Ex ante yield-spread specification

The paper summarizes the initial-yield regression as

$$\begin{aligned} \log(YieldSpread_{i,j,t}) = & \beta_1 IssuerShare_{k,t-1} + \gamma_1 (IssuerShare_{k,t-1} \times HOT_t) \\ & + \theta'_1 Rating_{i,j,t} + \theta'_2 Structure_{i,j,t} + \theta'_3 Collateral_{i,j,t} + \theta'_4 Issuer_{k,t} + \lambda_t + \varepsilon^1_{i,j,t}. \end{aligned} \quad (1)$$

- For AAA tranches, the main deal-structure variable is Fraction AAA.
- For non-AAA tranches, the main deal-structure variable is Level of Subordination.
- λ_t denotes cohort-year fixed effects.
- Some specifications add issuer fixed effects.

Interpretation. The key coefficient is γ_1 . A positive γ_1 means that large issuers pay higher yields in hotter markets, conditional on rating and controls. That is consistent with investors discounting the credibility of ratings from large issuers.

3.2 Ex post price-change specification

The paper summarizes the performance regression as

$$\begin{aligned} PriceChange_{i,j,t} = & \beta_2 IssuerShare_{k,t-1} + \gamma_2 (IssuerShare_{k,t-1} \times HOT_t) \\ & + \delta \log(YieldSpread_{i,j,t}) + \phi'_1 Rating_{i,j,t} + \phi'_2 Structure_{i,j,t} + \phi'_3 Collateral_{i,j,t} + \phi'_4 Issuer_{k,t} + \lambda_t + \varepsilon^2_{i,j,t}. \end{aligned} \quad (2)$$

Interpretation. A negative γ_2 means that large-issuer tranches perform worse in hot markets. Adding initial yield spreads tests whether issuance prices already captured that risk.

3.3 Economic meaning of the interaction

For a 10% market-share issuer in a hot year such as 2006, the yield effect from equation (1) is approximately

$$0.10\beta_1 + 0.10 \times HOT_{2006} \times \gamma_1.$$

Using the AAA specification without issuer fixed effects, the paper reports

$$-1.23 \times 0.10 + 9.8987 \times 0.25 \times 0.10 \approx 0.126,$$

which is about a 13% higher yield spread, or roughly 19 basis points.

Interpretation. The effect is economically meaningful because it is conditional on the stated rating. Investors are not simply distinguishing AAA from non-AAA; they are distinguishing which issuer brought the AAA tranche to market.

3.4 How the paper separates ex ante pricing from ex post performance

- The yield tests ask whether the market knew, or at least suspected, that large-issuer ratings were less reliable.
- The price-change tests ask whether large-issuer tranches actually did worse after the market turned.
- Including initial yield in the price-change regressions asks whether the ex ante price discount fully reflected the eventual performance gap.

Interpretation. Together, the two tests distinguish three possibilities: no ratings problem, a ratings problem fully priced at issuance, and a ratings problem only partly priced at issuance.

4 Identification and empirical design

4.1 What identifies the main effect

- The paper exploits cross-sectional differences in issuer size within issuance years.
- Issuance-year fixed effects absorb broad market conditions, including the housing boom and credit-spread environment.
- The interaction $IssuerShare \times HOT$ asks whether the size effect grows when MBS issuance is especially hot.
- Issuer fixed effects in some specifications compare a given issuer to itself across years as market share and HOT vary.

Interpretation. This is not a randomized design. Its credibility comes from showing that issuer-size pricing remains after controlling for ratings, deal structure, collateral composition, issuer quality, and time effects.

4.2 Why issuer size proxies for rating-agency conflicts

- Large issuers repeatedly bring deals to rating agencies.
- Repeated business gives them bargaining power and makes rating agencies more concerned about losing future fees.
- During boom years, short-run rating-agency revenue is especially high, while reputational costs may be delayed.
- Complex deals create disagreement across agencies, increasing the value of ratings shopping.

Interpretation. Issuer size is not itself a credit-quality characteristic. It is a proxy for the economic importance of an issuer to rating agencies.

4.3 Why the same-rating comparison matters

- Ratings are supposed to summarize credit quality.
- If ratings were equally informative across issuers, then two tranches with the same rating and similar controls should have similar yields.
- Higher yields on large-issuer tranches imply that investors believed the same rating carried more risk when the tranche was sold by a large issuer.

Interpretation. The paper’s title is answered by the yield regressions: the market behaved as though all ratings were not created equal.

4.4 Main limitations

- Issuer size is a proxy for rating-agency conflicts, not a direct observation of rating-agency bargaining.
- The paper observes purchased/reported ratings, not all preliminary ratings that issuers may have shopped.
- Initial yields may reflect liquidity, investor clienteles, or unobserved collateral features despite extensive controls.
- Price histories are available for a smaller subset of tranches than initial yields.
- The tests show that markets price the risk of inflated ratings; they do not directly observe the internal rating-agency decision process.

Interpretation. The evidence is strongest as a market-pricing test of perceived ratings inflation rather than as a direct audit of rating agencies’ private behavior.

5 Tables and figures

5.1 Table I: Summary statistics and ratings distribution

Read:

- The sample has 65,895 tranche observations with initial yield spreads and 9,299 observations with price changes.
- Mean initial yield spread is 147 bps; the median is 120 bps.
- Mean price change is -14.90%, while the median is -0.81%, showing severe left-tail losses.
- Fraction AAA averages 88.73% at the deal level, indicating that most private-label MBS financing is structured into highly rated tranches.
- Rating agencies rate roughly 60% of their tranches AAA, and AAA tranches account for a very large share of principal amount.
- Panel C compares big and small issuers across time periods and ratings groups, showing that large-issuer and small-issuer deals differ in ratings coverage, deal structure, and post-crisis performance.

Economic message. The MBS market is rating intensive and highly structured. Most dollars are financed through AAA claims, so small differences in rating credibility can have large market-wide consequences.

Table I
Summary Statistics of MBS and Issuer Characteristics

This table reports summary statistics of privately issued MBS sold between 2000 and 2006. *Fraction AAA* and *Number of Tranches in a Deal* are deal-level variables; all others vary at the tranche level. Variables are defined in Section II.B of the text. ***, **, and * in Panel C indicate significance at the 1%, 5%, and 10% levels, respectively, for either a two-sample *t*-test (on means) or a non-parametric median test (on medians) between characteristics of MBS sold by big and small issuers in a given time period.

Variable	<i>N</i>	Mean	Median	Std	p25	p75
Panel A: Overall Summary Statistics						
Fraction AAA (in%)	5,548	88.73	93.79	13.08	82.97	97.18
Initial yield spread (in bps)	65,895	147.07	120.00	447.63	51.49	185.00
Price change (in%)	9,299	-14.90	-0.81	28.36	-24.01	0.03
Issuer market share	85,272	0.05	0.04	0.04	0.03	0.07
Hot MBS market	86,635	0.17	0.19	0.06	0.11	0.25
Principal amount (in millions)	86,625	65.12	14.35	172.16	3.47	51.50
Weighted average life (in years)	70,484	5.63	4.90	3.36	3.28	7.26
Fra. of Colla. in troubled states (in%)	80,536	45.51	45.70	16.51	34.70	54.90
Herfindahl index of collateral	80,536	0.34	0.33	0.09	0.29	0.36
Initial rating	77,261	2.02	1.00	1.42	1.00	3.00
Issuer rating	77,198	2.90	2.67	0.91	2.44	3.11
Number of tranches in a deal	5,910	14.66	14.00	10.33	8.00	19.00
Level of subordination (in%)	83,655	93.00	97.00	12.00	93.00	99.00
Panel B: Description of Ratings Distribution						
Number of tranches in sample	Rating Agencies					
	Moody's	S&P	Fitch			
AAA	30,390	38,169	21,930			
Non-AAA	21,090	26,858	12,148			
Total	51,480	65,027	34,078			
Number Initial Ratings		Freq.	Percent			
1		15,031	19.45			
2		51,136	66.19			
3		11,094	14.36			
Total		77,261	100.00			
Same Originator Servicer		Freq.	Percent			
0 ("different")		14,872	35.49			
1 ("same")		27,034	64.51			
Total		41,906	100.00			
Rating disagreement		Freq.	Percent			
0 ("same")		54,063	86.88			
1 ("different")		8,167	13.12			
Total		62,230	100.00			

(Continued)

Table I—Continued

Panel B: Description of Ratings Distribution							
Types of Issuers		Freq.			Percent		
Commercial Banks		33,670			38.86		
Investment Banks		19,077			22.02		
Thriffs		17,388			20.07		
Finance Companies		8,013			9.25		
Others		8,487			9.80		
Total		86,635			100.00		
Deals with Floating Tranches		Freq.			Percent		
0 ("no")		1,626			27.51		
1 ("yes")		4,284			72.49		
Total		5,910			100.00		
Panel C: Distribution by Issuance Year and Issuer Size							
		2000–2003			2004–2006		
		Small	Big	Diff	Small	Big	Diff
Principal amount (in 000s)	<i>N</i>	7,835	18,605		25,164	35,021	
	Mean	68.06	54.15	13.91***	74.44	63.59	10.85***
	Median	17.50	12.00	5.50***	16.56	12.75	3.81***
	Std	158.07	133.53		206.51	165.76	
Weighted average life (in years)	<i>N</i>	6,804	15,829		19,916	27,935	
	Mean	5.90	6.68	-0.78***	4.99	5.44	-0.45***
	Median	5.13	5.78	-0.65***	4.79	4.76	0.03*
	Std	3.53	3.97		2.6	3.28	
Fra. of Colla. in troubled states	<i>N</i>	7,027	17,436		23,114	32,959	
	Mean	41.58	44.63	-3.05***	45.20	47.04	-1.84***
	Median	43.50	46.4	-2.90***	44.30	46.40	-2.10***
	Std	18.08	15.72		17.11	15.93	
Herfindahl index of collateral	<i>N</i>	7,027	17,436		23,114	32,959	
	Mean	0.34	0.36	-0.02***	0.33	0.34	-0.01***
	Median	0.33	0.34	-0.01***	0.31	0.32	-0.01***
	Std	0.10	0.10		0.09	0.08	
Initial rating	<i>N</i>	6,852	16,352		22,834	31,223	
	Mean	1.87	1.71	0.16***	2.24	2.04	0.20***
	Median	1.00	1.00	0.00***	1.67	1.00	0.67***
	Std	1.30	1.24		1.46	1.47	
Issuer rating	<i>N</i>	5,797	18,605		17,766	35,030	
	Mean	2.75	2.85	-0.10***	2.74	3.02	-0.28***
	Median	2.50	2.89	-0.39***	2.44	2.67	-0.23***
	Std	1.02	0.67		0.95	0.96	
Fraction AAA	<i>N</i>	771	1,376		1,544	1,857	
	Mean	90.25	92.11	-1.86***	85.67	88.13	-2.46***
	Median	94.19	96.52	-2.33***	88.41	93.92	-5.51***
	Std	11.21	11.97		12.26	14.50	

(Continued)

Table 1: Table I, Panels A–B

Table 2: Table I, Panel C, first block

Table I—Continued

Panel C: Distribution by Issuance Year and Issuer Size							
		2000–2003			2004–2006		
		Small	Big	Diff	Small	Big	Diff
Number of tranches in a deal	<i>N</i>	849	1,523		1,595	1,943	
	Mean	9.23	12.22	−2.99***	15.78	18.03	−2.25***
	Median	7.00	9.00	−2.00***	15.00	16.00	−1.00***
	Std	8.93	10.96		7.85	10.74	
Level of subordination	<i>N</i>	7,260	17,587		24,687	34,121	
	Mean	94.49	94.57	−0.08	91.61	91.99	−0.38***
	Median	97.62	97.51	0.11***	95.72	96.57	−0.85***
	Std	8.50	10.31		11.12	13.55	

		Num Initial Ratings	2000–2003	2004–2006
Small issuer	1		16.70%	15.36%
	2		67.67%	62.37%
	3		15.63%	22.27%
	Total		100.00%	100.00%
Big issuer	1		21.90%	21.78%
	2		71.16%	66.05%
	3		6.94%	12.18%
	Total		100.00%	100.00%

		Same Originator Servicer	2000–2003	2004–2006
Small issuer	0 (“different”)		36.31%	56.57%
	1 (“same”)		63.69%	43.43%
	Total		100.00%	100.00%
Big issuer	0 (“different”)		17.54%	27.14%
	1 (“same”)		82.46%	72.86%
	Total		100.00%	100.00%

		Rating Disagreement	2000–2003	2004–2006
Small issuer	0 (“same”)		94.90%	78.84%
	1 (“different”)		5.10%	21.16%
	Total		100.00%	100.00%
Big issuer	0 (“same”)		96.61%	86.27%
	1 (“different”)		3.39%	13.73%
	Total		100.00%	100.00%

(Continued)

Table I—Continued

Panel C: Distribution by Issuance Year and Issuer Size				
Rating Disagreement	0 (“same”)		1 (“different”)	Total
	Small	82.50%	17.50%	100.00%
	Big	89.82%	10.18%	100.00%
χ^2 test of diff.	703.36	<i>p</i> -value = 0.00		

		Deals with Floating Tranches	2000–2003	2004–2006
Small issuer	0 (“no”)		29.68%	14.61%
	1 (“yes”)		70.32%	85.39%
	Total		100.00%	100.00%
Big issuer	0 (“no”)		46.68%	22.13%
	1 (“yes”)		53.32%	77.87%
	Total		100.00%	100.00%

an MBS tranche between issuance and April 2009 (or its payoff date).⁸ This sample is considerably smaller than the yield sample because Bloomberg only provides pricing history for the larger deals.⁹ About 45% of the 9,299 tranches for which we have information on pricing history are paid off early and before the crisis, so the median price drop is only 0.8% while the mean drop is about 15%.

Table 3: Table I, Panel C, second block

Table 4: Table I, Panel C, final block

5.2 Table II and Figure 1: Large issuers and initial yields

Read:

- Table II shows that the top 10 issuers account for a very large share of annual private-label MBS issuance.
- The top-10 share ranges from about 55% to 68% over 2000–2006.
- Large recurring issuers include Countrywide, General Motors, Lehman, Bear Stearns, Bank of America, Credit Suisse, and WaMu.
- Figure 1 shows that large-issuer tranches generally have higher median initial yield spreads, especially around the boom years.
- Yield spreads decline after 2003 as the market booms, but large-issuer spreads remain above small-issuer spreads.

Economic message. The issuer market is concentrated. This concentration is the economic foundation for the bargaining-power and conflict-of-interest hypothesis.

Table II
Top 10 Issuers of Privately Sold MBS (by Number of Deals)

This table shows the top 10 issuers of privately sold MBS sorted by the number of deals in each year of the sample period. The last row shows the total market share of the top 10 issuers in terms of the number of deals they issue in a given year.

Rank	2000	2001	2002	2003	2004	2005	2006
1	General Motors	General Motors	General Motors	General Motors	Countrywide	Countrywide	Countrywide
2	Countrywide	Credit Suisse	Countrywide	Countrywide	General Motors	General Motors	General Motors
3	Wells Fargo	Countrywide	WaMu	Lehman	Bear Stearns	Lehman	Bear Stearns
4	JP Morgan Chase	Wells Fargo	Credit Suisse	Credit Suisse	Lehman	Bear Stearns	Lehman
5	Bank of America	Lehman	Bear Stearns	Bear Stearns	Bank of America	Bank of America	IndyMac
6	Citigroup	Citigroup	Bank of America	WaMu	Credit Suisse	Credit Suisse	WaMu
7	GreenPoint	Bank of America	Bear Stearns	WaMu	UBS	Goldman Sachs	Goldman Sachs
8	Lehman	Bear Stearns	Wells Fargo	Wells Fargo	Morgan Stanley	Merrill Lynch	Citigroup
9	GE	JP Morgan Chase	Morgan Stanley	UBS	Wells Fargo	Morgan Stanley	Citigroup
10	Conseco	WaMu	UBS	JP Morgan Chase	Goldman Sachs	Morgan Stanley	Credit Suisse
Market share	55.40%	68.12%	66.62%	64.25%	60.07%	59.17%	59.45%

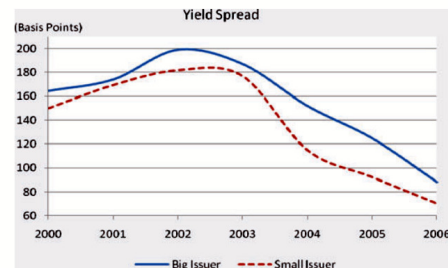


Table 5: Top 10 issuers by year

Figure 1: Initial yield spreads by issuer size

5.3 Table III: Yield-spread regressions

Read:

- Panel A estimates AAA-tranche regressions without issuer fixed effects.
- In Column 2 of Panel A, $HOT \times IssuerShare$ is strongly positive; the paper interprets the magnitude as roughly a 13% higher yield for a 10% market-share issuer in 2006.
- Adding subordination, collateral, rating-shopping, and complexity controls does not remove the issuer-size interaction.
- Panel B adds issuer fixed effects. The interaction remains statistically significant, and the implied effect is roughly a 10% higher yield for a 10% market-share issuer in a hot year.
- Panels C and D repeat the analysis for non-AAA tranches. The issuer-size effect remains and can be even larger with issuer fixed effects.
- Ratings-shopping variables matter more in the non-AAA market: fewer ratings and rating disagreement are associated with higher yields.

Economic message. The main result is not a simple comparison of risky and safe securities. Conditional on ratings and controls, the market charges higher yields to securities brought by large issuers during the boom.

Table III
Regression of MBS Yield Spread to Issuer Share

This table reports OLS regressions of the *Lag of Yield Spread* of privately issued MBS tranches on issuer market share, tranche-level characteristics, other issuer characteristics, and market conditions. The sample includes all tranches for which we can observe initial coupons (or fixed spreads) on Bloomberg originated between 2000 and 2006 that received a rating from Moody's, S&P, or Fitch. Variables are defined in Section II.B of the text. *Initial Rating Category Dummies* in Panels C and D are a set of dummies to indicate each level of the average ratings a given tranche received at issuance, after we convert the ratings into a numerical value by setting AAA = 1, AA+ = 1.67, AA = 2, AA- = 2.33, and so on, and then take the arithmetic averages of the ratings the tranche receives. Each regression includes separate intercepts for coupon types (floating, fixed, or variable). Standard errors are clustered by both cohort-year and issuer (two-way clustering). T-statistics are in parentheses. Panels A and B present results for AAA tranches only; Panels C and D present results for non-AAA tranches only. Panels A and C do not have issuer fixed effects, and Panels B and D have issuer fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: AAA Tranches Only; No Issuer Fixed Effects								
Issuer Share	0.2891	-1.2301**	0.2247	-1.2073**	0.1904	-1.1000**	0.2060	-1.1207**
	(0.53)	(-2.35)	(0.43)	(-2.21)	(0.36)	(-2.24)	(0.40)	(-2.12)
$HOT \times Issuer Share$	-	9.5987***	-	9.3310***	-	9.6203***	-	8.6405***
	-	(3.20)	-	(2.97)	-	(2.87)	-	(2.81)
Bank and Thrift	-0.0163	-0.0407	-0.0429	-0.0457	-0.0457	-0.0462	-0.0472	-0.0472
	(-0.35)	(-0.91)	(-1.09)	(-1.46)	(-1.54)	(-1.17)	(-1.21)	(-1.21)
Bank and Thrift Post July 04	0.0962**	0.1116***	0.1417***	0.1559***	0.1431***	0.1569***	0.1446***	0.1576***
	(2.20)	(2.61)	(3.27)	(3.56)	(3.25)	(3.42)	(3.16)	(3.27)
Level of Subordination	-	-	0.6736***	0.6718***	0.6645***	0.6629***	0.2417	0.2925
	-	-	(4.79)	(4.71)	(4.51)	(4.26)	(0.55)	(1.54)
$HOT \times Level of Subordination$	-	-	-	-	-	2.3559	2.1199	-
	-	-	-	-	-	(1.47)	(1.21)	-
Log of Principal	-0.0298**	-0.0210**	-0.0041	-0.0044	-0.0028	-0.0032	-0.0026	-0.0026
	(-2.20)	(-2.48)	(-0.46)	(-0.49)	(-0.32)	(-0.35)	(-0.24)	(-0.28)
Log of Weighted Average Life	0.0082	0.0081	0.0105	0.0108	0.0108	0.0110	0.0108	0.0108
	(0.08)	(0.08)	(0.11)	(0.11)	(0.11)	(0.11)	(0.11)	(0.11)
Log of Number of Tranches	-	-	0.1499***	0.1491***	0.1519***	0.1505***	0.1523***	0.1513***
	-	-	(5.32)	(4.88)	(5.12)	(4.70)	(5.14)	(4.80)

Table 6: Table III, Panel A, part 1

Table III—Continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: AAA Tranches Only; No Issuer Fixed Effects								
Fra. of Coll. in Troubled States	0.0043*	0.0042**	0.0031**	0.0031**	0.0029**	0.0029**	0.0029**	0.0029**
	(2.57)	(2.56)	(2.26)	(2.16)	(2.03)	(1.94)	(2.05)	(1.95)
Herdability Index of Collateral	-0.3241	-0.3277	-0.3056*	-0.3091*	-0.2841*	-0.2885*	-0.2870*	-0.2909*
	(-1.61)	(-1.60)	(-1.52)	(-1.78)	(-1.71)	(-1.67)	(-1.74)	(-1.71)
Same Originator Servicer	0.0921**	0.0892**	0.0713**	0.0660**	0.0712**	0.0660**	0.0679**	0.0680**
	(3.3)	(2.83)	(3.19)	(3.06)	(3.02)	(2.97)	(2.92)	(2.86)
Missing Originator Servicer	0.0031	0.0050	0.0030	0.0272	0.0270	0.0245	0.0265	0.0241
Issuer Rating	(1.56)	(1.39)	(1.09)	(0.98)	(0.94)	(0.87)	(0.86)	(0.89)
	(0.53)	(0.26)	(1.08)	(0.82)	(1.17)	(0.90)	(1.17)	(0.92)
One Initial Rating	-	-	-	-	0.0945	0.0891	0.0940	0.0889
	-	-	-	-	(1.57)	(1.47)	(1.58)	(1.52)
Two Initial Ratings	-	-	-	-	0.0775	0.0749	0.0770	0.0745
	-	-	-	-	(1.06)	(1.03)	(1.05)	(1.03)
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,129	25,129	25,129	25,129	25,129	25,129	25,129	25,129
R ²	0.17	0.18	0.729	0.729	0.729	0.729	0.729	0.729
Joint Wald tests of "One Initial Rating" and "Two Initial Ratings" (p-value)	-	-	-	-	3.43	3.80	4.59	4.52
	-	-	-	-	(0.18)	(0.15)	(0.10)	(0.10)

Table 7: Table III, Panel A, part 2

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: AAA Tranches Only, with Issuer Fixed Effects								
Issuer Share	0.0988	0.3832	0.3982	0.0167	0.4240	0.0737	0.3096	0.0496
<i>HOT</i> * Issuer Share	-	0.741	0.931	0.055	1.090	0.331	0.725	0.131
	-	2.629***	-	3.169***	-	2.903**	-	2.1836
	-	(2.72)	-	(3.10)	-	(1.71)	-	(1.16)
Bank and Thrift * Post July 04	0.1397***	0.1411***	0.1717***	0.1735***	0.1730***	0.1747***	0.1769***	0.1772***
	(2.75)	(2.53)	(3.02)	(2.45)	(3.01)	(2.39)	(3.01)	(2.35)
Level of Subordination	-	-	0.6468***	0.6468***	0.6463***	0.6472***	0.6359	0.6285
	-	-	(3.94)	(3.85)	(3.72)	(3.56)	(0.73)	(0.72)
<i>HOT</i> * Level of Subordination	-	-	-	-	-	-	2.3951	2.2242
	-	-	-	-	-	-	(1.32)	(1.26)
Log of Principal	-0.0188**	-0.0188**	-0.0036	-0.0035	-0.0026	-0.0026	-0.0021	-0.0021
	(-2.34)	(-2.14)	(-0.46)	(-0.44)	(-0.30)	(-0.32)	(-0.26)	(-0.25)
Log of Weighted Average Life	0.0104	0.0104	0.0120	0.0119	0.0123	0.0122	0.0124	0.0123
	(0.10)	(0.09)	(0.12)	(0.11)	(0.12)	(0.12)	(0.12)	(0.12)
Log of Number of Tranches	-	-	0.1297***	0.1397***	0.1406***	0.1406***	0.1418***	0.1418***
	-	-	(3.64)	(4.93)	(5.42)	(4.85)	(5.56)	(5.61)
Fra. of Colls. in Troubled States	0.0028	0.0028	0.0021	0.0022	0.0020	0.0020	0.0020	0.0020
	(1.63)	(1.63)	(1.27)	(1.35)	(1.24)	(1.26)	(1.26)	(1.28)
Herfindahl Index of Collateral	-0.1974	-0.2012	-0.1905	-0.1982	-0.1767	-0.1813	-0.1802	-0.1836
	(-0.92)	(-0.94)	(-1.03)	(-0.98)	(-0.93)	(-0.90)	(-0.97)	(-0.94)
Same Originator Servicer	0.1099***	0.1088***	0.0912***	0.0899***	0.0897***	0.0895***	0.0875***	0.0867***
	(3.35)	(3.31)	(6.29)	(5.13)	(6.42)	(5.17)	(4.43)	(4.76)
Missing Originator Servicer	0.0054*	0.0053*	0.0432	0.0431	0.0411	0.0411	0.0403	0.0403
	(1.56)	(1.79)	(1.46)	(1.45)	(1.26)	(1.33)	(1.38)	(1.40)
Issuer Rating	0.0774***	0.0749***	0.0691***	0.0649***	0.0731***	0.0692***	0.0697***	0.0668***
	(3.30)	(2.76)	(2.85)	(2.69)	(3.02)	(2.72)	(2.84)	(2.61)
One Initial Rating	-	-	-	-	0.0729	0.0707	0.0711	0.0703
	-	-	-	-	(1.00)	(0.98)	(1.00)	(0.99)

(Continued)

Table 8: Table III, Panel B, part 1

Panel C: Non-AAA Tranches Only, No Issuer Fixed Effects								
Fra. of Colls. in Troubled States	0.0010	0.0010	0.0008	0.0007	0.0007	0.0007	0.0007	0.0007
	(0.94)	(0.94)	(0.87)	(0.80)	(0.72)	(0.54)	(0.55)	(0.53)
Herfindahl Index of Collateral	0.0198	0.0078	0.0004	-0.0128	-0.0013	-0.0155	-0.0023	-0.0185
	(0.19)	(0.08)	(0.00)	(-0.12)	(-0.01)	(-0.13)	(-0.02)	(-0.16)
Same Originator Servicer	-0.0347	-0.0422*	-0.0339	-0.0410**	-0.0397	-0.0397	-0.0314	-0.0397
	(-1.60)	(-1.86)	(-1.53)	(-2.02)	(-1.41)	(-1.85)	(-1.42)	(-1.83)
Missing Originator Servicer	-0.0433***	-0.0454***	-0.0454***	-0.0444***	-0.0399***	-0.0414***	-0.0389***	-0.0417***
	(-2.33)	(-2.34)	(-2.74)	(-2.61)	(-2.60)	(-2.47)	(-2.64)	(-2.54)
Issuer Rating	0.0194	0.0175	0.0187	0.0167	0.0172	0.0152	0.0174	0.0153
	(1.00)	(1.20)	(1.56)	(1.17)	(1.54)	(1.30)	(1.54)	(1.35)
One Initial Rating	-	-	-	-	0.0669	0.0676	0.0666	0.0673
	-	-	-	-	(1.46)	(1.50)	(1.44)	(1.49)
Two Initial Ratings	-	-	-	-	0.0480***	0.0438**	0.0483***	0.0441**
	-	-	-	-	(2.70)	(2.17)	(2.60)	(2.43)
Rating Disagreement	-	-	-	-	0.1055***	0.1077***	0.1051***	0.1073***
	-	-	-	-	(4.40)	(4.57)	(4.36)	(4.46)
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Initial Rating Category Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,179	19,179	19,133	19,133	19,133	19,133	19,133	19,133
<i>R</i> ²	0.602	0.603	0.605	0.606	0.609	0.610	0.610	0.611
Joint Wald tests of "One Initial Rating" and "Two Initial Rating" (<i>p</i> -value)	-	-	-	-	8.30***	6.73***	8.00***	7.56***
	-	-	-	-	(0.02)	(0.03)	(0.02)	(0.02)
Panel D: Non-AAA Tranches Only, with Issuer Fixed Effects								
Issuer Share	0.0609	0.2688	0.8724	0.1397	0.8600	0.0503	0.7970	0.0282
	(1.64)	(0.53)	(1.51)	(0.36)	(1.50)	(0.14)	(1.46)	(0.07)
<i>HOT</i> * Issuer Share	-	5.919***	-	6.2108***	-	6.8432***	-	6.5064***
	-	(2.83)	-	(2.59)	-	(4.20)	-	(3.71)
Bank and Thrift * Post July 04	0.1717***	0.1732***	0.1884***	0.1801***	0.1880***	0.1897***	0.1913***	0.1928***
	(2.90)	(3.11)	(3.32)	(3.61)	(3.43)	(3.82)	(3.44)	(3.86)

(Continued)

Table 10: Table III, Panel C

5.4 Table IV: Split-sample yield evidence

Read:

- Panel A splits the sample into large versus small issuers and AAA versus non-AAA tranches.
- Level of subordination raises yields in both large-issuer and small-issuer subsamples.
- Bank/thrift issuance after July 2004 is associated with higher yields, consistent with regulatory-arbitrage concerns.
- Multiple ratings matter more for small issuers; investors can infer more from missing ratings when the issuer has less bargaining power.
- Panel B compares actual large-issuer yields to yields predicted using small-issuer coefficients.
- In 2004–2006, AAA tranches sold by large issuers average 98 bps, compared with 83 bps predicted under small-issuer pricing, an 18% increase.

Economic message. The split-sample evidence shows that the large-issuer yield premium is not only a linear-control artifact. It survives when the entire pricing model is allowed to differ for large and small issuers.

Table III—Continued								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: AAA Tranches Only, with Issuer Fixed Effects								
Two Initial Ratings	-	-	-	-	0.0562	0.0556	0.0551	0.0547
	-	-	-	-	(0.70)	(0.69)	(0.69)	(0.68)
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Issuer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,129	25,129	25,129	25,129	25,129	25,129	25,129	25,129
<i>R</i> ²	0.735	0.735	0.735	0.735	0.735	0.736	0.736	0.736
Joint Wald tests of "One Initial Rating" and "Two Initial Rating" (<i>p</i> -value)	-	-	-	-	1.76	1.64	1.71	1.79
	-	-	-	-	(0.42)	(0.44)	(0.43)	(0.41)
Panel C: Non-AAA Tranches Only, No Issuer Fixed Effects								
Issuer Share	0.3102	-1.0447***	0.3385	-1.0788***	0.3516	-1.1417***	0.3649	-1.1966***
	(1.00)	(-2.58)	(1.22)	(-2.58)	(1.08)	(-2.77)	(1.15)	(-2.81)
<i>HOT</i> * Issuer Share	-	7.9147***	-	8.3619***	-	8.7421***	-	8.7311***
	-	(4.51)	-	(4.43)	-	(4.92)	-	(4.85)
Bank and Thrift	-0.1179*	-0.1299**	-0.1196**	-0.1320**	-0.1226*	-0.1356**	-0.1238*	-0.1368**
	(-1.96)	(-2.09)	(-2.06)	(-2.43)	(-2.17)	(-2.36)	(-2.18)	(-2.35)
Bank and Thrift * Post July 04	0.1469***	0.1759***	0.1828***	0.1925***	0.1814***	0.1910***	0.1849***	0.1866***
	(2.83)	(2.91)	(3.16)	(3.70)	(3.27)	(3.43)	(3.34)	(3.81)
Level of Subordination	-	-	0.2921***	0.2979***	0.2764***	0.2820***	-0.1496	-0.1427
	-	-	(7.21)	(7.27)	(7.34)	(7.68)	(-0.80)	(-0.75)
<i>HOT</i> * Level of Subordination	-	-	-	-	-	-	2.3227*	2.2292*
	-	-	-	-	-	-	(1.75)	(1.72)
Log of Principal	0.0395*	0.0324	0.0331*	0.0349	0.0424**	0.0422**	0.0427**	0.0425**
	(1.66)	(1.56)	(1.71)	(1.71)	(2.56)	(2.42)	(2.68)	(2.22)
Log of Weighted Average Life	-0.0021	-0.0035	-0.0025	-0.0042*	-0.0075	-0.0069	-0.0082	-0.0096
	(-1.42)	(-1.42)	(-1.54)	(-1.68)	(-1.58)	(-1.69)	(-1.58)	(-1.59)
Log of Number of Tranches	-	-	0.0262	0.0268	0.0193	0.0198	0.0208	0.0213
	-	-	(1.08)	(1.12)	(0.87)	(0.92)	(0.95)	(0.97)

Table 9: Table III, Panel B, part 2

Table III—Continued								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel D: Non-AAA Tranches Only, with Issuer Fixed Effects								
Level of Subordination	-	-	0.3056***	0.3066***	0.2826***	0.2937***	-0.0995	-0.0620
	-	-	(4.26)	(4.17)	(4.17)	(4.22)	(-0.53)	(-0.41)
<i>HOT</i> * Level of Subordination	-	-	-	-	-	-	2.0903	1.9734
	-	-	-	-	-	-	(1.56)	(1.51)
Log of Principal	0.0309*	0.0302*	0.0331*	0.0333*	0.0409**	0.0412**	0.0412**	0.0414**
	(1.67)	(1.73)	(1.75)	(1.88)	(2.36)	(2.34)	(2.33)	(2.01)
Log of Weighted Average Life	-0.0067	-0.0068	-0.0044*	-0.0046	-0.0033*	-0.0039*	-0.0054*	-0.0059*
	(-1.57)	(-1.56)	(-1.68)	(-1.64)	(-1.71)	(-1.79)	(-1.56)	(-1.79)
Log of Number of Tranches	-	-	0.0121	0.0125	0.0094	0.0098	0.0039	0.0041
	-	-	(0.44)	(0.42)	(0.06)	(0.10)	(0.15)	(0.13)
Fra. of Colls. in Troubled States	0.0002	0.0002	0.0009	0.0001	-0.0002	-0.0001	-0.0002	-0.0001
	(0.13)	(0.17)	(0.63)	(0.07)	(-0.13)	(-0.08)	(-0.13)	(-0.08)
Herfindahl Index of Collateral	0.0977	0.0821	0.0841	0.0868	0.0876	0.0706	0.0857	0.0665
	(9.51)	(7.74)	(8.77)	(8.61)	(8.84)	(6.67)	(8.82)	(6.60)
Same Originator Servicer	0.0001	-0.0020	0.0019	-0.0002	-0.0008	-0.0022	-0.0012	-0.0024
	(0.01)	(-0.16)	(0.21)	(-0.02)	(-0.08)	(-0.29)	(-0.11)	(-0.32)
Missing Originator Servicer	-0.0238	-0.0235	-0.0209	-0.0206	-0.0220	-0.0217	-0.0225	-0.0222
	(-1.42)	(-1.38)	(-1.41)	(-1.33)	(-1.37)	(-1.35)	(-1.48)	(-1.22)
Issuer Rating	0.0028***	0.0588***	0.0593***	0.0549***	0.0578***	0.0530***	0.0556***	0.0519***
	(2.29)	(3.49)	(3.55)	(3.27)	(3.75)	(3.29)	(3.64)	(3.28)
One Initial Rating	-	-	-	-	0.0895**	0.0903**	0.0888**	0.0896**
	-	-	-	-	(2.18)	(2.24)	(2.07)	(2.03)
Two Initial Ratings	-	-	-	-	0.0415***	0.0415***	0.0415***	0.0397***
	-	-	-	-	(2.94)	(1.94)	(2.05)	(1.90)
Rating Disagreement	-	-	-	-	0.1104***	0.1118***	0.1088***	0.1111***
	-	-	-	-	(4.20)	(4.35)	(4.22)	(4.22)
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Issuer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Initial Rating Category Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,179	19,179	19,133	19,133	19,133	19,133	19,133	19,133
<i>R</i> ²	0.613	0.613	0.616	0.616	0.620	0.621	0.621	0.621
Joint Wald tests of "One Initial Rating" and "Two Initial Rating" (<i>p</i> -value)	-	-	-	-	7.89*	7.53*	7.79*	7.26*
	-	-	-	-	(0.02)	(0.02)	(0.02)	(0.03)

Table 11: Table III, Panel D

Table IV
Regression of MBS Yield Spread to Tranche Characteristics on
Samples Split by Issuer Size

This table reports OLS regressions of the *Log of Yield Spread* on tranche-level characteristics, various issuer characteristics, and market conditions for samples split by issuer market share. “Big issuer” means that the market share of the issuer falls into the top 10% of the market share distribution in a given year, and “Small issuers” refers to the rest of issuers in that year. Each regression includes separate intercepts for coupon types (floating, fixed, or variable). Standard errors are clustered by both cohort-year and issuer (two-way clustering). *T*-statistics are in parentheses. Panel A presents regression results for different subsamples and Panel B compares the average yield spread for tranches actually sold by big issuers with the predicted yield spread for those same tranches using the small-issuer coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Regression Coefficients				
	AAA Tranches		Non-AAA Tranches	
	Big issuers (1)	Small issuers (2)	Big issuers (3)	Small issuers (4)
Bank and Thrift	0.0114 (0.45)	−0.2011*** (−3.92)	−0.0580 (−0.93)	−0.2323*** (−3.83)
Bank and Thrift * Post July 04	0.0806* (1.68)	0.3237*** (6.58)	0.1217* (1.65)	0.3007*** (6.14)
Level of Subordination	0.7159*** (3.54)	0.6100*** (6.88)	0.2957*** (3.88)	0.2210*** (3.97)
Log of Principal	−0.0022 (−0.23)	−0.0065 (−0.49)	0.0465** (2.25)	0.0348 (1.64)
Log of Weighted Average Life	−0.0082 (−0.08)	0.0388 (0.36)	−0.0876 (−1.58)	−0.0907 (−1.21)
Log of Number of Tranches	0.1321*** (5.12)	0.1718*** (4.69)	−0.0091 (−0.35)	0.0573 (1.51)
Fra. of Colla. in Troubled States	0.0042* (1.89)	0.0009 (0.74)	0.0009 (0.71)	−0.0010 (−0.56)
Herfindahl Index of Collateral	−0.3479 (−1.43)	−0.1239 (−0.97)	−0.0270 (−0.17)	0.1052 (0.68)
Same Originator Servicer	0.0859*** (2.93)	0.0008 (0.02)	−0.0122 (−0.52)	−0.0979** (−2.53)
Missing Originator Servicer	0.0372 (0.82)	0.0141 (0.66)	−0.0420 (−1.53)	−0.0222 (−1.00)
Issuer Rating	0.0224 (1.13)	−0.0022 (−0.12)	0.0008 (0.05)	0.0239 (1.41)
One Initial Rating	0.0331 (0.44)	0.1876*** (3.04)	0.0964 (1.64)	0.0912 (1.63)
Two Initial Ratings	0.0267 (0.34)	0.1515*** (2.71)	0.0662* (1.80)	0.0431** (2.20)
Ratings disagreement	—	—	0.0979*** (3.18)	0.1190*** (5.66)
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes
Initial Rating Dummies	—	—	Yes	Yes
Observations	15,622	9,507	11,380	7,753
<i>R</i> ²	0.735	0.727	0.602	0.633

(Continued)

Table IV—Continued

Panel B: Actual and Predicted Mean Basis-Point Spreads for Large-Issuer Tranches (Predicted Values Based on Small-Issuer Coefficients)

Year	AAA Tranches		Non-AAA Tranches	
	Actual	Predicted	Actual	Predicted
2000	(1)	(2)	(3)	(4)
2001				
2002				
2003				
2004				
2005				
2006				
2007				
2008				
2009				
2010				
2011				
2012				
2013				
2014				
2015				
2016				
2017				
2018				
2019				
2020				

Table 12: Table IV, Panel A

Table 13: Table IV, Panel B

5.5 Figure 2 and Table V: Ex post price performance

Read:

- Figure 2 plots monthly prices for big-issuer and small-issuer tranches.
- For 2000–2003 cohorts, both groups are stable until the crisis period and then fall sharply.
- For 2004–2006 cohorts, big-issuer tranches fall earlier and more sharply than small-issuer tranches.
- Table V tests the same pattern with controls.
- In the baseline price-change regressions, large-issuer tranches from hot years experience larger price declines.
- The paper reports an implied effect of about 10 percentage points more price decline in the AAA market and about 11 percentage points more in the non-AAA market for a 10% market-share issuer during a hot year.

- Adding initial yield spreads attenuates the issuer-size effect only partly, suggesting that initial pricing captured some but not all of the ex post risk.

Economic message. The market recognized some risk at issuance, but ex post outcomes show that large-issuer boom-era tranches were still worse performers. The rating problem was only partially priced.

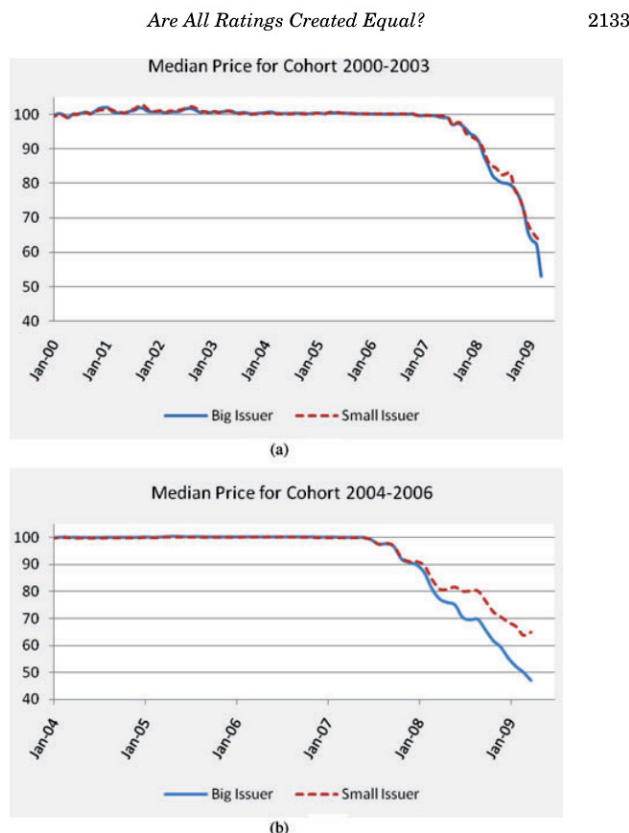


Figure 2. Monthly price changes of privately sold MBS (sorted by issuer market share and issuing year). This figure shows the change in the monthly price of privately issued MBS tranches sorted by issuer market share, calculated as the number of deals sold by the issuer divided by the total number of deals in the current year. “Big issuer” means that the market share of the issuer falls into the top 10% of the market share distribution in that year, and “Small issuer”

Figure 2: Figure 2: monthly price changes by issuer size and cohort

Table V
Regression of MBS Price Change to Issuer Share

This table reports OLS regressions of the change in the price of privately issued MBS tranches on issuer market share, other issuer and tranche-level characteristics and market conditions. The sample includes all tranches for which we can observe prices on Bloomberg originated between 2000 and 2006 that received at least one rating from Moody's, S&P, or Fitch. The dependent variable is the percentage change in the price of a tranche between issuance and April 2009 (or its payoff date). See Section II.B for other variable definitions. Each regression includes separate intercepts for coupon types (floating, fixed, or variable). Standard errors are clustered by both cohort year and issuer (two-way clustering). t -statistics are in parentheses. Panels A and B present results for AAA tranches only. Panels C and D show results for non-AAA tranches only. Panels A and C do not have issuer fixed effects, and Panels B and D have issuer fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Panel A: AAA Tranches Only, No Issuer Fixed Effects						
Issuer Share	-0.3707 (-1.60)	0.2802 (1.58)	0.4709** (2.27)	0.4747** (2.36)	0.3979** (2.15)	0.4024** (2.50)	0.3402 (0.95)
<i>HOT</i> * Issuer Share	-	-3.2553** (-2.12)	-5.8520*** (-2.73)	-5.7721*** (-2.80)	-5.2698** (-2.74)	-3.2683** (-2.59)	-
Bank and Thrift	-0.0087 (-0.78)	0.0049 (0.36)	0.0055 (0.43)	0.0075 (0.51)	0.0087 (0.59)	0.0051 (0.39)	0.0043 (0.30)
Bank and Thrift * Post July 04	0.0737* (1.89)	0.0630* (1.86)	0.0504* (1.66)	0.0444* (1.60)	0.0434* (1.57)	0.0428 (1.51)	0.0291 (0.85)
Level of Subordination	-	-	-0.1309** (-1.85)	-0.1349** (-1.86)	-0.0328 (-0.47)	-0.0997 (-1.34)	-0.0713 (-1.05)
<i>HOT</i> * Level of Subordination	-	-	-	-	-0.0736 (-1.04)	-0.2901 (-0.92)	-0.4069 (-0.90)
Log of Yield Spread	-	-	-	-	-	-	-0.0096 (-0.20)
<i>HOT</i> * Log of Yield Spread	-	-	-	-	-	-	-0.5516*** (-4.20)
Log of Principal	0.0002 (0.02)	-0.0011 (-0.11)	-0.0034 (-0.35)	-0.0017 (-0.19)	-0.0019 (-0.21)	0.0035 (0.39)	-0.0013 (-0.15)
Log of Weighted Average Life	-0.1400*** (-4.73)	-0.1480*** (-4.68)	-0.1494*** (-4.58)	-0.1480*** (-4.40)	-0.1405*** (-4.40)	-0.1494*** (-4.50)	-0.1594*** (-4.50)
Log of Number of Tranches	-	-	-0.0057 (-1.01)	-0.0078 (-1.09)	-0.0073 (-1.09)	-0.0024 (-0.34)	-0.0073 (-1.19)
Frm. of Coll. in Troubled States	-0.0002 (-0.22)	-0.0002 (-0.16)	-0.0001 (-0.15)	-0.0001 (-0.15)	-0.0002 (-0.12)	-0.0002 (-0.14)	-0.0002 (-0.15)
Hertfindahl Index of Collateral	0.0641 (0.52)	0.0706 (0.59)	0.0888 (0.78)	0.0877 (0.76)	0.0837 (0.71)	0.0859 (0.59)	0.0758 (0.60)

(Continued)

Table 14: Table V, Panel A, part 1

Table V—Continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Panel A: AAA Tranches Only, No Issuer Fixed Effects						
Same Originator Servicer	-0.0564** (-2.20)	-0.0474 (-1.53)	-0.0440 (-1.49)	-0.0408 (-1.31)	-0.0402 (-1.29)	-0.0401 (-1.32)	-0.0325 (-1.02)
Missing Originator Servicer	0.0146 (0.56)	0.0124 (0.45)	0.0125 (0.47)	0.0115 (0.43)	0.0112 (0.43)	0.0108 (0.37)	0.0046 (0.17)
Issuer Rating	-0.0050 (-0.97)	0.0008 (0.22)	0.0008 (0.20)	0.0017 (0.44)	0.0023 (0.46)	-0.0013 (-0.19)	-0.0045 (-0.64)
One Initial Rating	-	-	-	-0.0028 (-1.40)	-0.0096 (-1.32)	-0.0087 (-1.61)	-0.0045 (-1.13)
Two Initial Ratings	-	-	-	-0.0142 (-0.79)	-0.0142 (-0.78)	-0.0235 (-1.22)	-0.0194 (-1.01)
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,002	3,002	3,002	3,002	3,002	3,002	3,002
<i>R</i> ²	0.475	0.480	0.484	0.486	0.487	0.485	0.522
	Panel B: AAA Tranches Only, with Issuer Fixed Effects						
Issuer Share	0.0202 (0.07)	0.1349 (0.37)	0.2144 (0.53)	0.1932 (0.53)	0.1768 (0.49)	0.0295 (0.14)	-0.0766 (-0.29)
<i>HOT</i> * Issuer Share	-	-	-1.4895 (-0.89)	-1.8266 (-0.87)	-1.8566 (-0.85)	0.0290 (0.02)	1.3946 (0.87)
Bank and Thrift * Post July 04	0.0625 (1.34)	0.0634 (1.35)	0.0511 (1.15)	0.0497 (1.17)	0.0497 (1.15)	0.0409 (0.97)	0.0442 (1.07)
Level of Subordination	-	-	-0.1018 (-1.27)	-0.0916 (-0.91)	-0.0916 (-0.91)	-0.0464 (-1.52)	-0.0464 (-0.92)
<i>HOT</i> * Level of Subordination	-	-	-	-	-	-0.1669 (-0.30)	-0.2293 (-0.73)
Log of Yield Spread	-	-	-	-	-	-	-
Log of Principal	-	-	-	-	-	-	-
Log of Weighted Average Life	-	-	-	-	-	-	-
Log of Number of Tranches	-	-	-	-	-	-	-
Log of Principal	0.0001 (0.01)	0.0000 (0.00)	0.0004 (0.02)	0.0002 (0.01)	0.0002 (0.01)	0.0000 (0.01)	0.0002 (0.02)
Log of Weighted Average Life	-0.1502*** (-4.43)	-0.1501*** (-4.44)	-0.1499*** (-4.23)	-0.1501*** (-4.25)	-0.1501*** (-4.11)	-0.1489*** (-3.87)	-0.0660*** (-2.60)

Table 15: Table V, Panel A, part 2

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel B: AAA Tranches Only, with Issuer Fixed Effects							
Log of Number of Tranches	-	-	-0.0079	-0.0090	-0.0088	-0.0106	-0.0156
Prn. of Coll. in Troubled States	0.0005	0.0005	(-0.51)	(-0.56)	(-0.54)	(-0.52)	(-0.60)
Herfindahl Index of Collateral	0.1125	0.1149	0.0005	0.0005	0.0007	0.0004	0.0004
Same Originator Servicer	0.83	0.84	(0.87)	(0.86)	(0.81)	(0.75)	(0.89)
Missing Originator Servicer	(-0.139)	(-0.0136)	(-0.0132)	(-0.0119)	(-0.0115)	(-0.0095)	(-0.0025)
Issuer Rating	0.0065	0.0063	0.0003	0.0008	0.0005	0.0061	0.0045
One Initial Rating	(-0.46)	(-0.46)	(-0.44)	(-0.47)	(-0.52)	(-0.21)	(-0.077)
Two Initial Ratings	-	-	-	-	-	-	-
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Issuer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,602	3,602	3,602	3,602	3,602	3,605	3,605
R ²	0.514	0.515	0.516	0.517	0.517	0.531	0.555
Panel C: Non-AAA Tranches Only, No Issuer Fixed Effects							
Issuer Share	-0.0092	1.1876***	1.1896***	1.2137***	1.1607***	0.3390	0.3908
HOT* Issuer Share	(-0.15)	(0.41)	(0.60)	(0.60)	(0.70)	(1.03)	(1.03)
Bank and Thrift	-0.0238	-0.0165	-0.0164	-0.0157	-0.0157	-0.0029	0.0001
Bank and Thrift* Post July 04	(-0.86)	(-0.67)	(-0.63)	(-0.64)	(-0.64)	(-0.14)	(0.00)
Level of Subordination	-0.0445	-0.0507	-0.0483	-0.0484	-0.0469	-0.0087**	-0.0596**
HOT* Level of Subordination	(-1.21)	(-1.52)	(-1.41)	(-1.45)	(-1.37)	(-2.22)	(-2.31)
Log of Yield Spread	-	-	-	-	-	-	-
HOT* Log of Yield Spread	-	-	-	-	-	-	-
Log of Principal	-0.0178	-0.0165	-0.0161	-0.0140	-0.0138	-0.0171	-0.0185
Log of Weighted Average Life	(-1.59)	(-1.47)	(-1.51)	(-1.29)	(-1.24)	(-1.25)	(-1.45)
Log of Number of Tranches	-	-	-0.0171	-0.0156	-0.0157	0.0023	0.0034
Prn. of Coll. in Troubled States	0.0024**	0.0023**	0.0023**	0.0023**	0.0023**	0.0016	0.0016
	(2.10)	(2.08)	(2.09)	(2.14)	(2.02)	(1.27)	(1.31)

(Continued)

Table 16: Table V, Panel B, part 1

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel C: Non-AAA Tranches Only, No Issuer Fixed Effects							
Log of Yield Spread	-	-	-	-	-	-	0.0241**
HOT* Log of Yield Spread	-	-	-	-	-	-	(2.14)
Log of Principal	-0.0205*	-0.0218*	-0.0203*	-0.0169	-0.0187	-0.0284*	-0.0285**
Log of Weighted Average Life	(-1.75)	(-1.75)	(-1.66)	(-1.50)	(-1.46)	(-1.86)	(-2.17)
Log of Number of Tranches	-0.0171**	-0.0162**	-0.0159**	-0.0158**	-0.0156**	-0.0153**	-0.0157**
	(-2.70)	(-2.72)	(-2.67)	(-2.51)	(-2.52)	(-2.66)	(-2.57)
Prn. of Coll. in Troubled States	0.0012	0.0013	0.0013	0.0012	0.0012	0.0012	0.0012
Herfindahl Index of Collateral	(-0.223)	(-0.268)	(-0.307)	(-0.261)	(-0.197)	(-0.177)	(-0.175)
Same Originator Servicer	(-1.12)	(-1.10)	(-1.07)	(-1.04)	(-1.02)	(-0.96)	(-1.15)
Missing Originator Servicer	(-0.0487*)	(-0.0419*)	(-0.0437*)	(-0.0451**)	(-0.0451**)	(-0.0257)	(-0.0249)
Issuer Rating	(-1.89)	(-1.91)	(-2.00)	(-2.11)	(-2.17)	(-1.28)	(-1.30)
One Initial Rating	(-0.008)	(-0.007)	(-0.003)	(-0.0042)	(-0.0042)	(-0.0194)	(-0.018)
Two Initial Ratings	(-0.29)	(-0.32)	(-0.35)	(-0.40)	(-0.40)	(-1.01)	(-1.15)
Rating Disagreement	-	-	-	-	-	-	-
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Initial Rating Category Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,624	2,614	2,614	2,614	2,614	2,620	2,620
R ²	0.610	0.616	0.616	0.617	0.618	0.563	0.572

Table 17: Table V, Panel B, part 2

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel D: Non-AAA Tranches Only, with Issuer Fixed Effects							
Issuer Share	-0.3682	0.5423	0.5750	0.6779*	0.6283	-0.4480	-0.5414
HOT* Issuer Share	(-0.84)	(1.46)	(1.56)	(1.75)	(1.50)	(-0.86)	(-1.12)
Bank and Thrift	-0.0600	-0.0582	-0.0580	-0.0609*	-0.0582	-0.0079*	-0.0585*
Bank and Thrift* Post July 04	(-1.95)	(-1.64)	(-1.59)	(-1.47)	(-1.40)	(-2.26)	(-2.25)
Level of Subordination	-	-	-0.0157	-0.0127	-0.0122	-0.0168	-0.0232
HOT* Level of Subordination	-	-	(-0.31)	(-0.24)	(-0.48)	(-1.12)	(-1.63)
Log of Yield Spread	-	-	-	-	-	-	-
HOT* Log of Yield Spread	-	-	-	-	-	-	-
Log of Principal	-0.0178	-0.0165	-0.0161	-0.0140	-0.0138	-0.0171	-0.0185
Log of Weighted Average Life	(-1.59)	(-1.47)	(-1.51)	(-1.29)	(-1.24)	(-1.25)	(-1.45)
Log of Number of Tranches	-	-	-0.0171	-0.0156	-0.0157	0.0023	0.0034
Prn. of Coll. in Troubled States	0.0024**	0.0023**	0.0023**	0.0023**	0.0023**	0.0016	0.0016
	(2.10)	(2.08)	(2.09)	(2.14)	(2.02)	(1.27)	(1.31)

(Continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel D: Non-AAA Tranches Only, with Issuer Fixed Effects							
Herfindahl Index of Collateral	-0.0015	-0.0015	-0.0015	-0.0015	-0.0015	-0.0015	-0.0015
Same Originator Servicer	-0.0239	-0.0211	-0.0203	-0.0207	-0.0220	-0.0096	-0.0071
Missing Originator Servicer	(-0.90)	(-1.56)	(-1.80)	(-1.99)	(-1.64)	(-0.23)	(-0.37)
Issuer Rating	(-1.33)	(-1.12)	(-1.16)	(-1.09)	(-1.04)	(-0.66)	(-0.75)
One Initial Rating	(-0.000)	(-0.000)	(-0.000)	(-0.000)	(-0.000)	(-0.000)	(-0.000)
Two Initial Ratings	(-0.000)	(-0.000)	(-0.000)	(-0.000)	(-0.000)	(-0.000)	(-0.000)
Rating Disagreement	-	-	-	-	-	-	-
Cohort-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Initial Rating Category Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,624	2,614	2,614	2,614	2,614	2,620	2,620
R ²	0.642	0.646	0.646	0.646	0.647	0.603	0.611

Table 19: Table V, Panel D

Table 18: Table V, Panel C

6 Short summary

- The paper studies whether private-label MBS ratings were equally informative across issuers.
- The key conflict is the issuer-pays rating model: large issuers repeatedly bring deals and can threaten rating-agency fee revenue.
- The sample covers non-GSE MBS issued from 2000 to 2006 and price histories through April 2009.
- The key treatment-like variable is issuer market share, interacted with hot MBS market conditions.
- Initial yield spreads are higher for tranches sold by large issuers during boom years, even conditional on ratings.
- The result applies for both AAA and non-AAA tranches.
- The effect remains after controls for collateral, deal structure, ratings shopping, issuer type, issuer rating, and issuer fixed effects.
- MBS sold by banks and thrifts after July 2004 carry higher yields, consistent with regulatory-arbitrage incentives.
- More complex deals have higher yields, suggesting investor concern about structured-product complexity.
- Multiple ratings and rating disagreement are particularly informative in non-AAA tranches.
- Ex post prices fall more for large-issuer boom-era tranches.
- Initial yields partly, but not fully, capture the later underperformance.

- The broad message is that the same credit rating did not mean the same risk across issuers.

7 Conclusion

7.1 Core conclusion

He, Qian, and Strahan show that investors priced MBS ratings as issuer-dependent signals. Tranches sold by large issuers, especially during the 2004–2006 boom, carried higher initial yields than similarly rated tranches sold by small issuers. These same large-issuer securities also performed worse after issuance.

7.2 Economic intuition

Large issuers generated repeated fee business for rating agencies. This made conflicts of interest more severe under the issuer-pays model. Investors understood that ratings from large issuers could be more inflated and demanded higher yields. But the later price declines show that the market did not fully price the risk.

7.3 Takeaway

The paper’s central lesson is that ratings are not pure, issuer-invariant measures of credit quality. In structured finance, ratings reflect not only collateral risk but also the incentives of issuers, rating agencies, and regulated investors. The market recognized this problem during the boom, but only incompletely.